AS · Unit 1 · WCH11 | Structure, Bonding and Introduction to Organic Chemistry

Formulae, equations and amounts of substance

Atoms are far too small to count one at a time, so chemists count them by weighing, by measuring gas volumes and by measuring solution volumes. This topic builds the idea of the mole, the three equations that link it to things you can measure, and the balanced equations that tell you how much of each substance reacts and forms. Almost every calculation in A level chemistry rests on what is here.

SPEC 1.1 Atoms, elements, ions, molecules, compounds and formulae

WHAT THIS SPEC POINT ASKS

Know what is meant by atom, element, ion, molecule, compound, empirical formula and molecular formula.

The words you must keep apart

Term	Meaning	Examples
Atom	the smallest particle of an element that keeps its chemical identity; neutral (protons = electrons)	Na, C, Ar
Element	a substance made of only one type of atom (all atoms have the same number of protons)	sodium, oxygen (O_2), ozone (O_3), diamond
Ion	an atom or group of atoms carrying an overall charge, because electrons have been lost or gained	Na^+, Cl^-, NH_4^+, SO_4^{2-}
Molecule	two or more atoms held together by covalent bonds, as a discrete particle	O_2, H_2O, CH_4
Compound	a substance made of two or more elements chemically bonded together	H_2O, NaCl, CO_2
Empirical formula	the **simplest whole-number ratio** of atoms of each element in a compound	CH_2O for glucose
Molecular formula	the **actual number** of atoms of each element in one molecule	$C_6H_{12}O_6$ for glucose

Quick sort: which words fit?

Species	Element or compound?	Particle type
Ne	element	atom
O_2 and O_3	element (two different molecules of the same element)	molecule
H_2O	compound	molecule
NaCl	compound	**not** a molecule: a giant ionic lattice; NaCl is its formula unit
SiO_2	compound	**not** a molecule: a giant covalent lattice
NH_4^+	part of a compound	ion (a polyatomic ion)

Empirical vs molecular formula

Substance	Molecular formula	Empirical formula
ethane	C_2H_6	CH_3
butane	C_4H_{10}	C_2H_5
benzene	C_6H_6	CH
glucose	$C_6H_{12}O_6$	CH_2O
water	H_2O	H_2O (same)

- Divide the subscripts by their highest common factor to get the empirical formula.
- Giant structures (ionic or giant covalent) have **only** an empirical formula: NaCl, SiO_2, MgO. There is no "molecule" to count.

EXAM TIP

Counting hydrogens in a skeletal formula one by one is slow and error-prone. Count the carbons, start from the alkane formula C_nH_{2n+2}, then **subtract 2 H for every C=C and for every ring**. Don't forget the carbons in methyl branches.

MARK-SCHEME WORDING

- Molecular formula = "the actual number of atoms of each element"; empirical formula = "the simplest whole-number ratio".

- For a giant structure say "relative **formula** mass", not relative molecular mass.

Often asked as: MCQs on formulae and particles; give both the empirical and the molecular formula of an alkene or alkane; work out a molecular formula from a skeletal formula.

COMMON MISTAKE

Giving the molecular formula when the **empirical** formula is asked for (or the other way round). Calling NaCl or SiO_2 a "molecule". Mixing up "atom" and "ion" when counting electrons or charges.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Working out the molecular formula of a big natural molecule (rings plus several C=C bonds) from its skeletal formula (WCH11 Jun 2022).
- Using the alkane general formula to get the formula of a large alkane such as squalane (WCH11 Jan 2021).

SPEC 1.2 The mole and the Avogadro constant

WHAT THIS SPEC POINT ASKS

Know that the mole is the unit of amount of substance, and use the Avogadro constant (6.02×10^{23} mol^{-1}) in calculations.

What a mole is

- A **mole** is the amount of substance that contains **6.02×10^{23} particles**. This number is the **Avogadro constant**, L (or N_A), units **mol^{-1}**.
- It is the number of atoms in exactly 12 g of carbon-12. So 1 mol of any substance has a mass in grams equal to its relative mass (molar mass, 1.4).
- The "particles" can be atoms, molecules, ions, electrons or formula units. **Always say which.**

The three links you use all the time

Want	Use	Units
moles from mass	$n = m \div M$	m in g, M in g mol^{-1}

Want	Use	Units
number of particles from moles	$N = n \times L$	$L = 6.02 \times 10^{23}$ mol^{-1}
moles from particles	$n = N \div L$	

The route is always **mass** ⇄ **moles** ⇄ **particles**. Go through moles.

WORKED EXAMPLE

Worked example: atoms in a mass of a gas

How many **atoms** are in 5.60 g of nitrogen gas, N_2?

Moles of N_2 = 5.60 ÷ 28.0 = 0.200 mol

Each N_2 molecule has 2 atoms, so moles of N atoms = 0.400 mol

Atoms = 0.400 × 6.02 × 10^{23} = **2.41 × 10^{23} atoms**

(If the question asked for molecules, the answer would be 1.20 × 10^{23}.)

WORKED EXAMPLE

Worked example: molecules from tonnes

How many molecules are in 1.00 tonne of ethene, C_2H_4 (M = 28.0 g mol^{-1})?

1.00 tonne = 1.00 × 10^6 g

Moles = 1.00 × 10^6 ÷ 28.0 = 3.57 × 10^4 mol

Molecules = 3.57 × 10^4 × 6.02 × 10^{23} = **2.15 × 10^{28} molecules**

Don't multiply by 6 here: 6 would give atoms, and the question asked for molecules.

WORKED EXAMPLE

Worked example: counting ions and electrons

(a) 17.1 g of aluminium sulfate, $Al_2(SO_4)_3$ (M = 342.3 g mol^{-1}). Moles = 17.1 ÷ 342.3 = 0.0500 mol.

Each formula unit holds 2 Al^{3+} + 3 SO_4^{2-} = 5 ions, so ions = 5 × 0.0500 × 6.02 × 10^{23} = **1.50 × 10^{23} ions**.

(b) 1.00 g of Mg^{2+} ions. Moles = 1.00 ÷ 24.3 = 0.0412 mol. Each Mg^{2+} has 12 − 2 = 10 electrons.

Electrons = 10 × 0.0412 × 6.02 × 10^{23} = **2.48 × 10^{23} electrons**.

(c) Mass of one carbon-12 atom = 12.0 ÷ 6.02 × 10^{23} = **1.99 × 10^{-23} g**.

EXAM TIP

Huge and tiny numbers are normal here. Use the EXP or ×10ˣ key, keep the full calculator value between steps, and round only the final answer (usually 3 s.f.).

Counting atoms from a mass

1. How many atoms are there in 5.00 g of aluminium? (A_r: Al = 27.0)
2. Find the amount: $n = \frac{m}{M} = \frac{5.00}{27.0} = 0.1852 \text{ mol}$

   Keep the unrounded value in your calculator for the next step.
3. Multiply by the Avogadro constant: $N = 0.1852 \times 6.02 \times 10^{23} = 1.11 \times 10^{23}$ atoms

Atoms inside molecules

1. How many atoms are there in 0.500 mol of ammonia, NH_3?
2. Number of molecules: $0.500 \times 6.02 \times 10^{23} = 3.01 \times 10^{23}$
3. Each molecule contains 4 atoms (one N and three H), so the number of atoms is $4 \times 3.01 \times 10^{23} = 1.20 \times 10^{24}$.

   The same idea gives the mass of a single atom: one carbon-12 atom has a mass of $12 \div (6.02 \times 10^{23}) = 1.99 \times 10^{-23}$ *g.*

MARK-SCHEME WORDING

- Show moles first, then multiply by 6.02 × 10²³.
- Count **atoms** when the question says atoms (×2 for N_2, ×6 for C_2H_4 and so on), and **molecules** when it says molecules.

Often asked as: MCQs on numbers of particles or formula units; atoms in a mass of gas; molecules in a mass given in tonnes or kg; molecules allowed in a room (with ppm).

COMMON MISTAKE

Mixing up atoms and molecules, which halves or doubles the answer. An extra factor of 6 when counting molecules of ethene. Dividing by *L* when you should multiply.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Unit conversions given in the stem: tonnes → kg → g (WCH11 Jan 2020), hectares → km² (WCH11 Jan 2021).
- Choosing between formula units and individual ions in an MCQ (WCH11 Jan 2026).

SPEC 1.3 Balanced full and ionic equations with state symbols

WHAT THIS SPEC POINT ASKS

Write balanced full equations and ionic equations, with state symbols, for chemical reactions.

Balancing a full equation

1. Write the correct formula for every reactant and product (use the ion charges below for ionic compounds).
2. Count the atoms of each element on both sides.
3. Balance by changing the **numbers in front** (coefficients). **Never change a subscript**: that changes the substance.
4. For combustion, balance **C first, then H, then O** last. A fraction such as 11/2 O_2 is allowed (or double everything).
5. Add state symbols: **(s)** solid, **(l)** liquid, **(g)** gas, **(aq)** dissolved in water.

WORKED EXAMPLE

Worked example: combustion

Complete combustion of heptane (a liquid):

C: 7 C → $7CO_2$. H: 16 H → $8H_2O$. O on the right = 14 + 8 = 22 atoms = 11 O_2.

$C_7H_{16}(l) + 11O_2(g) \rightarrow 7CO_2(g) + 8H_2O(l)$

Dodecane: $C_{12}H_{26}(l) + 18½O_2(g) \rightarrow 12CO_2(g) + 13H_2O(l)$ (or double it: $2C_{12}H_{26} + 37O_2 \rightarrow 24CO_2 + 26H_2O$).

Ions you need to know

Positive ions	Negative ions
H^+, Na^+, K^+, Li^+, **Ag^+**, NH_4^+	OH^-, Cl^-, Br^-, I^-, NO_3^-, HCO_3^-
Mg^{2+}, Ca^{2+}, Ba^{2+}, Zn^{2+}, Cu^{2+}, Fe^{2+}, Pb^{2+}	O^{2-}, S^{2-}, SO_4^{2-}, CO_3^{2-}
Al^{3+}, Fe^{3+}, Cr^{3+}	PO_4^{3-}, N^{3-}

To write an ionic formula, balance the charges: Al^{3+} and SO_4^{2-} need 2 × (+3) = 3 × (−2), so **$Al_2(SO_4)_3$**. Use brackets around a polyatomic ion when you need more than one.

Writing an ionic equation

1. Write the balanced full equation with state symbols.
2. Split every **aqueous ionic compound** (and strong acid) into its ions. Leave solids, liquids (including water), gases and covalent molecules as they are.
3. Cancel the ions that appear unchanged on both sides (**spectator ions**).
4. Check that atoms **and charges** balance.

WORKED EXAMPLE

Worked example: ionic equations

Precipitation. $AgNO_3(aq) + NaCl(aq) \rightarrow AgCl(s) + NaNO_3(aq)$

Split: $Ag^+ + NO_3^- + Na^+ + Cl^- \rightarrow AgCl(s) + Na^+ + NO_3^-$. Na^+ and NO_3^- are spectators.

$Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$

Acid + carbonate. $BaCO_3(s) + 2HCl(aq) \rightarrow BaCl_2(aq) + CO_2(g) + H_2O(l)$

$BaCO_3$ is a solid, so it stays whole. Cl^- is the spectator.

$BaCO_3(s) + 2H^+(aq) \rightarrow Ba^{2+}(aq) + CO_2(g) + H_2O(l)$

Charge check: left +2, right +2.

EXAM TIP

Work ionic equations out every time; don't try to recall them. Underlined words such as "including state symbols" in the stem mean the second mark is for states.

Writing an ionic equation

1. Start with the balanced full equation, including state symbols:
   $AgNO_3(aq) + NaCl(aq) \rightarrow AgCl(s) + NaNO_3(aq)$

2. Split every aqueous ionic compound into its ions:
   $Ag^+(aq) + NO_3^-(aq) + Na^+(aq) + Cl^-(aq) \rightarrow AgCl(s) + Na^+(aq) + NO_3^-(aq)$

   AgCl is a solid precipitate, so it stays together.

3. Cross out ions that appear unchanged on both sides (Na^+ and NO_3^-):
   $Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$

   Check the charges: 1+ and 1− on the left give zero overall, and so does the neutral solid on the right. Any soluble silver salt mixed with any soluble chloride gives this same equation, which is why ionic equations are so useful.

MARK-SCHEME WORDING

- Usually 2 marks: **M1** correct formulae and balancing; **M2** state symbols (only if M1 is correct or a near miss).
- "Allow multiples" of a balanced equation.
- In ionic equations, **cancel spectator ions**: "do not award uncancelled chloride ions" (WCH11 Oct 2022). One paper let uncancelled ions through, but don't rely on it.
- Acid + carbonate: the products are CO_2 + H_2O. "Do not award H_2CO_3 as a product".
- Water in a combustion equation may be (l) or (g).

Often asked as: combustion of an alkane; correct a wrong equation; build an ionic equation for a precipitation, displacement or acid reaction; MCQ "which is the ionic equation for…".

COMMON MISTAKE

- State symbols left out, even when the question underlines them. A liquid fuel such as heptane is **(l)**, not (aq). Ions in solution are **(aq)**, not (l).
- Halogens written as atoms: it is **Cl_2**, **Br_2**, not Cl or Br.
- Wrong ion charges (Ag^+, not Ag^{2+}; Cu^{2+} in copper(II) compounds). Learn the table above.
- Writing water as the product of a metal + acid. The gas is **hydrogen**.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Predicting formulae of unfamiliar products from ion charges, e.g. Cr_2O_3, $Cu(NO_3)_2$, Na_2HPO_4 (WCH11 Oct 2019).
- Balancing a decomposition: $2FeSO_4 \cdot 7H_2O \rightarrow Fe_2O_3 + SO_2 + SO_3 + 14H_2O$ (WCH11 Jun 2021).
- Equations for thermal and photo-decomposition, e.g. $2AgCl(s) \rightarrow 2Ag(s) + Cl_2(g)$ (WCH11 Jun 2024).

SPEC 1.4 Relative masses, molar mass and parts per million

WHAT THIS SPEC POINT ASKS

(i) Relative atomic mass on the ^{12}C scale. (ii) Relative molecular mass and relative formula mass, calculated from Ar values ("formula mass" for giant structures). (iii) Molar mass in g mol^{-1}. (iv) Parts per million (ppm), including gases in the atmosphere.

The ^{12}C scale

Atoms are far too light to weigh in grams, so their masses are compared with **one-twelfth of the mass of a carbon-12 atom**, which is defined as exactly 12.

Term	Definition (exam wording)
Relative isotopic mass	the mass of an atom of one isotope compared with 1/12 of the mass of an atom of carbon-12
Relative atomic mass, A_r	the **weighted mean (average) mass of an atom** of an element compared with **1/12 of the mass of an atom of carbon-12**
Relative molecular mass, M_r	the sum of the A_r values of all the atoms in one **molecule** (simple molecular substances)
Relative formula mass	the same sum for one **formula unit** of a giant structure (ionic or giant covalent: NaCl, MgO, SiO_2)
Molar mass, M	the mass of **one mole** of a substance, in **g mol^{-1}**; numerically equal to A_r or M_r

- A_r is a **weighted mean** because most elements are mixtures of isotopes (2.6 shows how to calculate it). That's why chlorine's A_r is 35.5, not a whole number.
- Relative masses have **no units**. Molar mass has units of g mol^{-1}.

WORKED EXAMPLE

Worked example: relative formula mass of a hydrate

$CuSO_4 \cdot 5H_2O$ (A_r: Cu 63.5, S 32.1, O 16.0, H 1.0)

$CuSO_4 = 63.5 + 32.1 + (4 \times 16.0) = 159.6$

$5H_2O = 5 \times 18.0 = 90.0$

Relative formula mass = 159.6 + 90.0 = **249.6** (molar mass 249.6 g mol^{-1})

The dot means "plus": add the water, don't multiply by it.

Parts per million (ppm)

- **ppm = parts per million**: the number of parts of a substance in every million (10^6) parts of the mixture.
- ppm = (amount of the substance ÷ total amount) × 10^6, using the **same units** top and bottom.
- For solutions and solids it is usually **by mass** (1 ppm = 1 mg per kg, or about 1 mg per dm^3 of water). For **gases in the atmosphere** it is **by volume** (1 ppm = 1 cm^3 of the gas in every 10^6 cm^3 = 1 m^3 of air).
- Useful conversion: 1% = 10 000 ppm. CO_2 in air at about 420 ppm is 0.042%.

WORKED EXAMPLE

Worked example: ppm

(a) A toothpaste contains 1.45 mg of fluoride in every 1.00 g of paste.

ppm = (1.45 × 10^{-3} g ÷ 1.00 g) × 10^6 = **1450 ppm**

(b) The safe limit of a gas is 1.1 ppm by volume. What volume of the gas can a 50.0 m^3 room hold?

Volume of gas = 50.0 × 1.1 ÷ 10^6 = 5.5 × 10^{-5} m^3 = **55 cm^3**

Check that the answer makes sense: a few tens of cm^3 in a whole room is plausible.

Amount of a hydrated salt

1. What amount of $CuSO_4 \cdot 5H_2O$ is in 2.50 g of the blue crystals?
2. $M = 63.5 + 32.1 + (4 \times 16.0) + (5 \times 18.0) = 249.6\ \text{g mol}^{-1}$

   Each H_2O has $M = 18.0$. Forgetting the water, and using 159.6, is the usual error.

3. $n = \dfrac{2.50}{249.6} = 0.0100 \text{ mol}$

Working with ppm

1. A 250 g sample of fish contains 0.100 mg of mercury. What is its mercury content in ppm by mass?
2. Use the same units for both masses: $0.100 \text{ mg} = 1.00 \times 10^{-4} \text{ g}$
3. $\text{ppm} = \dfrac{1.00 \times 10^{-4}}{250} \times 10^{6} = 0.400 \text{ ppm}$
4. The other way round: air contains about 420 ppm of carbon dioxide by volume, so 1.00 dm^3 of air holds $420 \times 10^{-6} \times 1.00 = 4.20 \times 10^{-4} \text{ dm}^3$, which is 0.420 cm^3 of CO_2.

MARK-SCHEME WORDING

- A_r definition (2 marks): **M1** "weighted mean mass / average mass of an **atom**" (the word *atom* must be there); **M2** "compared to 1/12 (the mass) of a ^{12}C atom". "Element" and "isotopes" are ignored.
- Give an A_r to the number of s.f. or d.p. asked: 35.49, not 35.4938.
- ppm: write out "parts per million", then set out as part ÷ whole × 10^6.

Often asked as: define relative atomic mass; calculate A_r from abundances (2.6); say what ppm means; ppm of fluoride in toothpaste, of a pollutant in a room, or a gas "mixing ratio".

COMMON MISTAKE

- Leaving out "mean", "carbon-12" or "1/12" in the A_r definition. Writing "mass of an element" instead of "mass of an atom".
- Saying "relative molecular mass" for an ionic compound: use relative **formula** mass.
- Getting a ppm answer that is obviously silly (a room volume of 10^{-9} dm^3) and not checking it.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- A weighted mean in a new context, e.g. the average property of a fuel blend such as E10 (WCH11 Oct 2023).
- Unfamiliar concentration units for gases, such as g m^{-3} (WCH11 Oct 2022).

SPEC 1.5 Concentration in mol dm^{-3} and g dm^{-3}

WHAT THIS SPEC POINT ASKS

Calculate the concentration of a solution in mol dm^{-3} and in g dm^{-3}. (Titration calculations come later.)

The equations

- **Concentration (mol dm^{-3}) = moles ÷ volume in dm^3**, so $n = c \times V$.
- **Concentration (g dm^{-3}) = mass ÷ volume in dm^3**.
- Convert between them with the molar mass: g dm^{-3} = mol dm^{-3} × M.
- **1 dm^3 = 1000 cm^3**. Divide cm^3 by 1000 to get dm^3 before you use these equations.

WORKED EXAMPLE

Worked example: making a solution

(a) 2.00 g of NaOH (M = 40.0) is dissolved to make 250 cm^3 of solution.

Moles = 2.00 ÷ 40.0 = 0.0500 mol. Volume = 250 ÷ 1000 = 0.250 dm^3.

c = 0.0500 ÷ 0.250 = **0.200 mol dm^{-3}**; in g dm^{-3}: 2.00 ÷ 0.250 = **8.00 g dm^{-3}**.

(b) What mass of $CuSO_4 \cdot 5H_2O$ (M = 249.6) makes 100 cm^3 of 0.150 mol dm^{-3} solution?

Moles = 0.150 × 0.100 = 0.0150 mol. Mass = 0.0150 × 249.6 = **3.74 g**.

Use the hydrate's molar mass because the hydrate is what you weigh.

WORKED EXAMPLE

Worked example: ion concentrations and dilution

(a) 0.100 mol dm^{-3} $Al_2(SO_4)_3$: [Al^{3+}] = 2 × 0.100 = **0.200 mol dm^{-3}**; [SO_4^{2-}] = 3 × 0.100 = **0.300 mol dm^{-3}**.

(b) 25.0 cm^3 of 2.00 mol dm^{-3} HCl is made up to 250.0 cm^3 with water. Moles don't change:

c = 2.00 × 25.0 ÷ 250.0 = **0.200 mol dm^{-3}**.

WORKED EXAMPLE

Worked example: minimum concentration of acid

0.200 g of $MgCO_3$ (M = 84.3) must react completely with 25.0 cm^3 of hydrochloric acid.

$MgCO_3 + 2HCl \rightarrow MgCl_2 + CO_2 + H_2O$

Moles of $MgCO_3$ = 0.200 ÷ 84.3 = 2.37×10^{-3} mol, so HCl needed = $2 \times 2.37 \times 10^{-3} = 4.74 \times 10^{-3}$ mol.

Minimum c = $4.74 \times 10^{-3} \div 0.0250$ = **0.190 mol dm⁻³**.

A "minimum" must not be rounded down: 0.18 would not be enough acid.

Concentration of a standard solution

1. 3.18 g of anhydrous sodium carbonate, Na_2CO_3, is dissolved in water and made up to 250.0 cm^3. Find its concentration in mol dm^{-3} and in g dm^{-3}.
2. $M = (2\times 23.0) + 12.0 + (3\times 16.0) = 106.0\ \text{g mol}^{-1}$, so $n = \dfrac{3.18}{106.0} = 0.0300\ \text{mol}$
3. Volume in dm^3: $250.0 \div 1000 = 0.2500\ \text{dm}^3$
4. $c = \dfrac{0.0300}{0.2500} = 0.120\ \text{mol dm}^{-3}$
5. In g dm^{-3}: $0.120 \times 106.0 = 12.7\ \text{g dm}^{-3}$

   Check: 3.18 g in a quarter of a cubic decimetre is the same as 12.7 g in a whole one.

Counting ions in a solution

1. How many chloride ions are in 25.0 cm^3 of 0.150 mol dm^{-3} calcium chloride solution?
2. $n(CaCl_2) = c \times V = 0.150 \times 0.0250 = 3.75 \times 10^{-3}\ \text{mol}$
3. Each $CaCl_2$ gives two Cl^- ions: $n(Cl^-) = 2 \times 3.75 \times 10^{-3} = 7.50 \times 10^{-3}\ \text{mol}$
4. $N = 7.50 \times 10^{-3} \times 6.02 \times 10^{23} = 4.52 \times 10^{21}$ ions

Diluting an acid

1. What volume of water must be added to 10.0 cm^3 of 2.00 mol dm^{-3} hydrochloric acid to dilute it to 0.250 mol dm^{-3}?

2. Amount of HCl, which does not change: $n = 2.00 \times 0.0100 = 0.0200 \text{ mol}$

3. Volume needed at the new concentration: $V = \frac{n}{c} = \frac{0.0200}{0.250} = 0.0800 \text{ dm}^3 = 80.0 \text{ cm}^3$

4. Water to add $= 80.0 - 10.0 = 70.0 \text{ cm}^3$

The concentration fell by a factor of 8, so the final volume must be 8 times the starting volume.

Concentration from a titration

1. 25.0 cm^3 of sodium hydroxide solution is titrated with 0.100 mol dm^{-3} sulfuric acid. The titres are 22.10 (rough), 21.65, 21.55 and 21.60 cm^3. Find the concentration of the sodium hydroxide.
$H_2SO_4(aq) + 2NaOH(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(l)$

2. Mean of the concordant titres: $(21.65 + 21.55 + 21.60) \div 3 = 21.60 \text{ cm}^3$

Leave out the rough titre. These three lie within 0.10 cm^3 of each other.

3. Amount of acid used: $n = 0.100 \times \frac{21.60}{1000} = 2.160 \times 10^{-3} \text{ mol}$

4. The ratio H_2SO_4 : NaOH is 1 : 2, so $n(\text{NaOH}) = 4.320 \times 10^{-3} \text{ mol}$

5. $c(\text{NaOH}) = \frac{4.320 \times 10^{-3}}{0.0250} = 0.173 \text{ mol dm}^{-3}$

MARK-SCHEME WORDING

- $n = c \times V$ with V in dm^3 ($\text{cm}^3 \div 1000$); g dm^{-3} = mass ÷ volume in dm^3.
- Use the mole ratio from the equation before converting.

Often asked as: MCQs on concentration or the mass to weigh out; the concentration of one ion; minimum concentration of acid in a practical context.

COMMON MISTAKE

- Leaving the volume in cm^3 but quoting the answer in mol dm^{-3} (a factor of 1000 out).
- Forgetting the 1 : 2 ratio between carbonate and HCl.

- Ion concentrations: forgetting that one formula unit can release 2 or 3 of an ion.

SPEC 1.6 Empirical and molecular formulae from experimental data

WHAT THIS SPEC POINT ASKS

Use experimental data (masses or percentages) to calculate empirical formulae and then molecular formulae.

The method (set it out as a table)

1. Write the **mass** (or %) of each element. If one element isn't given, find it **by difference** (often oxygen).
2. Divide each by its $\mathbf{A_r}$ (not its atomic number) to get moles of atoms.
3. Divide every answer by the **smallest** one.
4. If you get a number like 1.5, 1.33 or 1.25, **multiply all of them** by 2, 3 or 4 to get whole numbers. Don't round 1.5 to 2.
5. Write the empirical formula.
6. **Molecular formula**: work out the M_r of the empirical formula, divide the real M_r by it, and multiply the empirical formula by the answer.

WORKED EXAMPLE

Worked example: percentages, then molecular formula

A hydrocarbon is 92.3% carbon; M_r = 78.0.

	C	H
%	92.3	100 − 92.3 = 7.7
÷ A_r	92.3 ÷ 12.0 = 7.69	7.7 ÷ 1.0 = 7.7
÷ smallest	1	1

Empirical formula **CH** (M_r 13.0). 78.0 ÷ 13.0 = 6, so molecular formula **C_6H_6**.

WORKED EXAMPLE

Worked example: a ratio that isn't whole

An oxide of iron is 70.0% Fe and 30.0% O.

Fe: 70.0 ÷ 55.8 = 1.25; O: 30.0 ÷ 16.0 = 1.875.

Divide by the smallest: Fe 1, O 1.50. Multiply both by 2: Fe 2, O 3 → **Fe_2O_3**.

WORKED EXAMPLE

Worked example: oxygen by difference

A compound contains C 54.5%, H 9.1% and oxygen only. M_r = 88.0.

O = 100 − 54.5 − 9.1 = 36.4%.

C 54.5 ÷ 12.0 = 4.54; H 9.1 ÷ 1.0 = 9.1; O 36.4 ÷ 16.0 = 2.275 → ratio 2 : 4 : 1 → **C_2H_4O** (M_r 44.0).

88.0 ÷ 44.0 = 2, so **$C_4H_8O_2$**. Showing the 44.0 is what earns the last mark.

WORKED EXAMPLE

Worked example: water of crystallisation

2.46 g of hydrated magnesium sulfate, $MgSO_4 \cdot xH_2O$, is heated to constant mass. 1.20 g of anhydrous $MgSO_4$ (M = 120.4) is left.

Mass of water = 2.46 − 1.20 = 1.26 g → 1.26 ÷ 18.0 = 0.0700 mol

Moles of $MgSO_4$ = 1.20 ÷ 120.4 = 0.00997 mol

x = 0.0700 ÷ 0.00997 = 7.02 → **x = 7**, $MgSO_4 \cdot 7H_2O$. Waters of crystallisation are always whole numbers.

WORKED EXAMPLE

Worked example: from combustion products

0.840 g of a hydrocarbon burns to give 2.64 g CO_2 and 1.08 g H_2O. Its M_r is 84.0.

Moles of C = moles of CO_2 = 2.64 ÷ 44.0 = 0.0600 mol (mass of C = 0.720 g)

Moles of H = 2 × moles of H_2O = 2 × (1.08 ÷ 18.0) = 0.120 mol (mass of H = 0.120 g)

0.720 + 0.120 = 0.840 g, so there is no oxygen in the compound.

C : H = 0.0600 : 0.120 = 1 : 2 → **CH_2** (M_r 14.0). 84.0 ÷ 14.0 = 6 → **C_6H_{12}**.

Empirical and molecular formula from percentages

1. A hydrocarbon contains 85.7% carbon and 14.3% hydrogen by mass. Its M_r is 56.0. Find its molecular formula.

2. Take 100 g, so there are 85.7 g of C and 14.3 g of H. Divide by A_r: C: $85.7 \div 12.0 = 7.14$ mol; H: $14.3 \div 1.0 = 14.3$ mol

3. Divide by the smaller value: C: $7.14 \div 7.14 = 1$; H: $14.3 \div 7.14 = 2.00$. The empirical formula is CH_2.

4. Empirical formula mass $= 12.0 + 2.0 = 14.0$, and $56.0 \div 14.0 = 4$. The molecular formula is C_4H_8.

M_r does not need to be exact. The multiple must be a whole number, so a measured value of 55 or 57 still points to 4.

When the ratio is not a whole number

1. An oxide of iron contains 69.9% iron by mass. Find its empirical formula.

2. Oxygen makes up the rest: $100 - 69.9 = 30.1\%$. Fe: $69.9 \div 55.8 = 1.253$; O: $30.1 \div 16.0 = 1.881$

3. Divide by the smaller: Fe = 1, O = $1.881 \div 1.253 = 1.50$

4. 1.50 is not a whole number, so double both: Fe = 2, O = 3. The empirical formula is Fe_2O_3.

Rounding 1.50 to 2 would give FeO_2, which is wrong. Only round values that are within about 0.1 of a whole number.

Finding x in $MgSO_4 \cdot xH_2O$

1. 4.92 g of hydrated magnesium sulfate, $MgSO_4 \cdot xH_2O$, is heated to constant mass. 2.40 g of anhydrous $MgSO_4$ is left. Find x.

2. Mass of water lost $= 4.92 - 2.40 = 2.52$ g

3. $n(MgSO_4) = \dfrac{2.40}{120.4} = 0.01993$ mol and $n(H_2O) = \dfrac{2.52}{18.0} = 0.140$ mol

$M(MgSO_4) = 24.3 + 32.1 + 4 \times 16.0 = 120.4 \text{ g mol}^{-1}$.

4 Divide by the smaller amount: $\frac{0.140}{0.01993} = 7.02$, so $x = 7$ and the formula is $MgSO_4 \cdot 7H_2O$.

MARK-SCHEME WORDING

- Marks for: moles of each element (mass ÷ A_r), the simplest whole-number ratio, the formula. "Correct formula with no working scores 1 mark only."
- Molecular formula: show the M_r of the empirical formula and the multiplier.
- TE (error carried forward) is given if you use a wrong A_r but the method is right.

Often asked as: empirical formula from % composition (oxygen by difference); molecular formula from M_r (sometimes M_r comes from pV = nRT, 1.8); water of crystallisation from heating data.

COMMON MISTAKE

- Not calculating the oxygen at all; using atomic numbers instead of A_r.
- Rounding 1 : 1.5 to 1 : 2.
- Stopping at the empirical formula when the molecular formula was asked; not simplifying ($C_4H_8O_2$ is not an empirical formula).
- Calculating moles of O_2 when you need moles of O atoms.
- Swapping the masses of water and anhydrous salt in a hydrate question.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- The third element found by difference (WCH11 Jun 2024, Jan 2024).
- Ratios such as 1 : 1 : 1.5 that must be doubled (WCH11 Oct 2021).
- Formula of a hydrate from a % of the anhydrous salt, e.g. 78.5% anhydrous: use ×100 ÷ 78.5, not × 78.5 (WCH11 Oct 2019).

SPEC 1.7 Reacting masses

WHAT THIS SPEC POINT ASKS

Use balanced equations to calculate reacting masses (and work back from masses), using amount of substance and molar mass.

Three steps, every time

1. **Moles of what you know**: $n = m \div M$.
2. **Mole ratio** from the balanced equation: moles of what you want.
3. **Convert**: mass = $n \times M$ (or volume, or concentration).

The **mole** ratio is not the mass ratio. Never use the equation's numbers on masses directly.

WORKED EXAMPLE

Worked example: mass of product

What mass of calcium oxide forms when 25.0 g of calcium carbonate is heated?

$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$

Moles of $CaCO_3$ = 25.0 ÷ 100.1 = 0.250 mol

Ratio 1 : 1, so 0.250 mol of CaO

Mass = 0.250 × 56.1 = **14.0 g**

WORKED EXAMPLE

Worked example: limiting reagent

2.43 g of Mg is added to 50.0 cm^3 of 1.00 mol dm^{-3} HCl. What volume of hydrogen forms at r.t.p.?

$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$

Moles of Mg = 2.43 ÷ 24.3 = 0.100 mol. Moles of HCl = 1.00 × 0.0500 = 0.0500 mol.

0.100 mol of Mg would need 0.200 mol of HCl, but only 0.0500 mol is there. **HCl is limiting**; Mg is in excess.

Moles of H_2 = 0.0500 ÷ 2 = 0.0250 mol. Volume = 0.0250 × 24.0 = **0.600 dm^3**

Always calculate the product from the **limiting** reagent.

WORKED EXAMPLE

Worked example: identify an unknown metal

1.56 g of a Group 1 metal M reacts with water to give 0.480 dm^3 of hydrogen at r.t.p.

$2M(s) + 2H_2O(l) \rightarrow 2MOH(aq) + H_2(g)$

Moles of H_2 = 0.480 ÷ 24.0 = 0.0200 mol, so moles of M = 2 × 0.0200 = 0.0400 mol

A_r = 1.56 ÷ 0.0400 = 39.0 → **potassium**. Check the name you give matches the number you got.

WORKED EXAMPLE

Worked example: industrial scale

Aluminium is extracted by electrolysis: $2Al_2O_3 \rightarrow 4Al + 3O_2$. Mass of Al from 1.00 tonne of Al_2O_3 (M = 102.0)?

Moles of Al_2O_3 = $1.00 \times 10^6 \div 102.0 = 9.80 \times 10^3$ mol; moles of Al = $2 \times 9.80 \times 10^3 = 1.96 \times 10^4$ mol

Mass = $1.96 \times 10^4 \times 27.0 = 5.29 \times 10^5$ g = **0.529 tonnes**

If you keep the same mass unit on both sides (tonnes in, tonnes out) you can skip the conversion to grams.

Reacting masses and gas volume

1. 25.0 g of calcium carbonate is heated until it has fully decomposed. Find the mass of calcium oxide formed and the volume of carbon dioxide at r.t.p.
   $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$
2. Step 1: $n(\mathrm{CaCO_3}) = \dfrac{25.0}{100.1} = 0.2498 \text{ mol}$
3. Step 2: the ratio is 1 : 1 : 1, so $n(\mathrm{CaO}) = n(\mathrm{CO_2}) = 0.2498 \text{ mol}$
4. Step 3: $m(\mathrm{CaO}) = 0.2498 \times 56.1 = 14.0 \text{ g}$ and $V(\mathrm{CO_2}) = 0.2498 \times 24.0 = 5.99 \text{ dm}^3$

   Check with conservation of mass: CO_2 has mass $0.2498 \times 44.0 = 11.0$ g, and 14.0 + 11.0 = 25.0 g.

Finding the limiting reactant

1. 2.00 g of magnesium is added to 50.0 cm^3 of 1.00 mol dm^{-3} hydrochloric acid. What volume of hydrogen forms at r.t.p.?
2. $n(\mathrm{Mg}) = 2.00 \div 24.3 = 0.0823 \text{ mol}$ and $n(\mathrm{HCl}) = 1.00 \times 0.0500 = 0.0500 \text{ mol}$
3. Mg + 2HCl: 0.0500 mol of HCl reacts with only 0.0250 mol of Mg. There is 0.0823 mol of Mg, so magnesium is in excess and the acid is limiting.
4. $n(\mathrm{H_2}) = \frac{1}{2} \times 0.0500 = 0.0250 \text{ mol}$, so $V = 0.0250 \times 24.0 = 0.600 \text{ dm}^3 = 600 \text{ cm}^3$

   Using the magnesium would predict 1.98 dm^3, which is impossible: the acid has run out long before.

MARK-SCHEME WORDING

- Marked as a chain: moles of one substance; ratio from the equation; moles or mass of the other. Each step is a mark, with TE.
- "Correct answer with no working scores full marks" is common, but a wrong answer with no working scores nothing.

Often asked as: maximum (theoretical) mass of product; identify a metal or compound from its A_r or M_r; limiting reagent in a salt preparation.

COMMON MISTAKE

- Ignoring the mole ratio in the equation.
- Using the mass of a hydrate when the anhydrous mass is needed (or the reverse).
- Rounding intermediate values; working out 38.8 correctly then naming the wrong element.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Identifying an unknown metal or gas by calculating A_r or M_r (WCH11 Jan 2026, Oct 2023, Jun 2023).
- % by mass of an element in a compound, e.g. % nitrogen in a fertiliser = (mass of N in the formula ÷ M_r) × 100 (WCH11 Jan 2026).

SPEC 1.8 Gas volumes, molar volume and pV = nRT

WHAT THIS SPEC POINT ASKS

Use equations to calculate gas volumes (and back), using (i) amount of substance, (ii) the molar volume of a gas, (iii) pV = nRT for gases and volatile liquids.

(i) and (ii) Molar volume and Avogadro's law

- **Avogadro's law**: equal volumes of all gases at the same temperature and pressure contain the **same number of molecules**.
- So one mole of **any** gas has the same volume, the **molar volume**. At room temperature and pressure (r.t.p., about 298 K and 1 atm) it is **24.0 dm³ mol⁻¹** (24 000 cm³ mol⁻¹). Use the value the question gives.
- Volume of gas = moles × 24.0 dm³ mol⁻¹; moles = volume ÷ 24.0.
- For reactions between gases, **volume ratio = mole ratio**, so you can work with volumes directly.

WORKED EXAMPLE

Worked example: volume of gas from a mass

Volume of CO_2 at r.t.p. from 5.00 g of $CaCO_3$ reacting with excess acid?

$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$

Moles of $CaCO_3$ = 5.00 ÷ 100.1 = 0.0500 mol = moles of CO_2

Volume = 0.0500 × 24.0 = **1.20 dm³**

WORKED EXAMPLE

Worked example: gas volumes only

20 cm³ of propane is burned in 150 cm³ of oxygen. Volumes are measured at r.t.p., so the water is a liquid.

$C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(l)$

O_2 used = 5 × 20 = 100 cm³, so 50 cm³ of O_2 is left over. CO_2 made = 3 × 20 = 60 cm³.

Final gas volume = 60 + 50 = **110 cm³**. Passing it through NaOH(aq) absorbs the CO_2, leaving **50 cm³** (the excess O_2).

(iii) The ideal gas equation

pV = nRT

Symbol	Quantity	SI unit	Conversions
p	pressure	**Pa**	kPa × 10^3 → Pa
V	volume	**m³**	cm³ × 10^{-6} → m³; dm³ × 10^{-3} → m³
n	amount	mol	
R	gas constant	8.31 J mol^{-1} K^{-1}	given in the data booklet
T	temperature	**K**	°C + 273

It works for any gas, and for a **volatile liquid** once it has been vaporised (so you can find its M_r). From it, the molar volume at 298 K and 101 kPa is $V = RT \div p$ = 8.31 × 298 ÷ 101 000 ≈ 0.0245 m³, about 24 dm³.

WORKED EXAMPLE

Worked example: moles from p, V and T

How many moles of gas are in 250 cm³ at 101 kPa and 25 °C?

$p = 101 \times 10^3$ Pa; $V = 250 \times 10^{-6} = 2.50 \times 10^{-4}$ m³; $T = 25 + 273 = 298$ K

$n = pV \div RT = (1.01 \times 10^5 \times 2.50 \times 10^{-4}) \div (8.31 \times 298)$ = **0.0102 mol**

WORKED EXAMPLE

Worked example: molar mass of a volatile liquid

0.275 g of a liquid hydrocarbon is vaporised. The vapour occupies 98.0 cm³ at 100 °C and 1.01×10^5 Pa.

$n = (1.01 \times 10^5 \times 98.0 \times 10^{-6}) \div (8.31 \times 373) = 3.19 \times 10^{-3}$ mol

M = mass ÷ moles = $0.275 \div 3.19 \times 10^{-3}$ = **86.1 g mol^{-1}**

An alkane with M_r 86 is C_6H_{14} (hexane). Don't stop at n and call it the molar mass.

WORKED EXAMPLE

Worked example: counting C=C bonds

0.0100 mol of a hydrocarbon reacts completely with 480 cm³ of hydrogen at r.t.p. (Ni catalyst).

Moles of H_2 = 0.480 ÷ 24.0 = 0.0200 mol. Each C=C takes one H_2.

0.0200 ÷ 0.0100 = 2, so there are **2 C=C bonds** per molecule.

EXAM TIP

Write the converted values before you substitute: "$V = 2.50 \times 10^{-4}$ m³, $T = 298$ K". Each conversion is often a separate mark, and the volume to m³ is the mark most often lost.

Reacting gas volumes

1. 40 cm^3 of propane is burned in 250 cm^3 of oxygen. All volumes are measured at r.t.p. What is the total volume of gas at the end?
   $C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(l)$

2. Oxygen used: $5 \times 40 = 200\ cm^3$, so $250 - 200 = 50\ cm^3$ of oxygen is left over.

3. Carbon dioxide formed: $3 \times 40 = 120\ \text{cm}^3$

4. At r.t.p. the water is a liquid with a negligible volume, so the gas left is $50 + 120 = 170\ \text{cm}^3$.

Relative molecular mass of a volatile liquid

1. 0.200 g of a volatile liquid is injected into a gas syringe in an oven at 100 °C. It evaporates completely and the vapour occupies 78.0 cm^3 at 101 kPa. Find M_r.

2. Convert the units: $p = 101\,000\ \text{Pa}$, $V = 78.0 \times 10^{-6}\ \text{m}^3$, $T = 100 + 273 = 373\ \text{K}$

3. $$n = \frac{pV}{RT} = \frac{101\,000 \times 78.0 \times 10^{-6}}{8.31 \times 373} = 2.54 \times 10^{-3}\ \text{mol}$$

4. $M = \dfrac{m}{n} = \dfrac{0.200}{2.54 \times 10^{-3}} = 78.7\ \text{g mol}^{-1}$, so M_r = 78.7. The liquid could be benzene, C_6H_6 (M_r = 78.0).

Gas volume at other conditions

1. What volume of hydrogen, at 25 °C and 100 kPa, is made when 0.500 g of magnesium reacts with excess hydrochloric acid?
   Mg(s) + 2HCl(aq) → $MgCl_2$(aq) + H_2(g)

2. $n(\text{Mg}) = \dfrac{0.500}{24.3} = 0.02058\ \text{mol}$, and the ratio Mg : H_2 is 1 : 1.

3. $$V = \frac{nRT}{p} = \frac{0.02058 \times 8.31 \times 298}{100\,000} = 5.10 \times 10^{-4}\ \text{m}^3$$

4. $5.10 \times 10^{-4}\ \text{m}^3 \times 10^6 = 510\ \text{cm}^3$

MARK-SCHEME WORDING

- Usual structure: **M1** *V* in m³; **M2** *T* in K; **M3** rearranged equation; **M4** answer, sometimes a mark for 2 or 3 s.f.
- 1 dm³ = 10^{-3} m³; 1 cm³ = 10^{-6} m³; kPa × 10^3 = Pa.
- For gas-only reactions the volume-ratio method is accepted. If O_2 is in excess, add the unused O_2 to the gas left at the end.
- Molar volume at r.t.p.: "24 dm³ mol^{-1}".

Often asked as: moles from p, V, T; then use *n* to find M_r of a gas or volatile liquid, to count C=C bonds by hydrogenation, or to show that a vapour is a dimer (Al_2Cl_6); gas-volume ratios in combustion.

COMMON MISTAKE

- Not converting cm^3 to m^3, or multiplying by 10^6 instead of 10^{-6}. Converting cm^3 to dm^3 and leaving kPa gives the right number only by luck.
- Forgetting to add 273, or adding it twice.
- Rounding *n* to 1 s.f. (0.002437 → 0.002 is 18% out).
- Using 24 $dm^3\ mol^{-1}$ for a liquid or solid: molar volume is for gases only.
- Giving the answer in dm^3 when your working gives m^3.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Density of a gas instead of a volume: 1.00 dm^3 of a gas has mass 0.654 g at 298 K and 101 kPa → *n* = 0.0408 mol → *M* = 16.0, so methane (WCH11 Jan 2024).
- Liquid oxygen: 1.00 kg → 1000 ÷ **32.0** = 31.3 mol → $V = nRT \div p \approx 0.766\ m^3$ at 298 K and 101 kPa. Divide by 32, not 16 (WCH11 Jan 2023).
- Showing that aluminium chloride vapour is Al_2Cl_6 from its M_r (WCH11 Oct 2024).
- Predicting the volume change in a gas reaction (WCH11 Jan 2026).

SPEC 1.9 Percentage yield and atom economy

WHAT THIS SPEC POINT ASKS

Calculate percentage yields and percentage atom economies (by mass) for lab and industrial processes, from equations and experimental results.

Percentage yield

$$\%\ \text{yield} = (\text{actual mass of product} \div \text{theoretical mass}) \times 100$$

- The **theoretical** (maximum) yield comes from a reacting-mass calculation using the limiting reagent (1.7). You can use moles instead of masses.
- Yields are below 100% because: the reaction is **incomplete** or **reversible**; **side reactions** make other products; product is **lost** when transferring, filtering or purifying; reactants are impure.

- A "yield" over 100% means the product is **impure or still wet** (e.g. not dried to constant mass).

WORKED EXAMPLE

Worked example: percentage yield

Heating 25.0 g of $CaCO_3$ should give 14.0 g of CaO (1.7). A student gets 11.2 g.

% yield = 11.2 ÷ 14.0 × 100 = **79.9%** (use the unrounded 14.01 g)

Scaling up: to make 5.00 g of a product at 75.0% yield, plan for a theoretical yield of 5.00 × 100 ÷ 75.0 = 6.67 g. A yield below 100% means you need **more** reactant, not less.

Atom economy

> % atom economy = (molar mass of the desired product ÷ sum of the molar masses of all products) × 100

- Use the **balanced equation**, so multiply each molar mass by its coefficient.
- The sum of the products' masses equals the sum of the reactants' masses, so you get the same answer either way, but **use products**, as the spec does.
- Atom economy is a **theoretical** measure of how much of the starting material ends up in the useful product. It doesn't depend on the yield.

WORKED EXAMPLE

Worked example: atom economy

(a) Hydrogen from methane: $CH_4 + H_2O \rightarrow CO + 3H_2$

Desired: 3 × 2.0 = 6.0. All products: 28.0 + 6.0 = 34.0. Atom economy = 6.0 ÷ 34.0 × 100 = **17.6%**

(b) Iron: $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

Desired: 2 × 55.8 = 111.6. All products: 111.6 + 3 × 44.0 = 243.6. Atom economy = **45.8%**

(c) Ethanol: hydration $C_2H_4 + H_2O \rightarrow C_2H_5OH$ is **100%** (one product). Fermentation $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$ gives 92.0 ÷ 180.0 × 100 = **51.1%**.

Why high atom economy matters

- Less **waste** made and fewer raw materials used per kg of product: more **sustainable**, and cheaper.
- Addition reactions (and addition polymerisation) have **100%** atom economy; substitution and elimination reactions always make a by-product.

- If the by-product can be **sold or used**, the process is less wasteful than its atom economy suggests.
- A process needs **both** a high yield and a high atom economy to be efficient.

Yield and atom economy for making lime

1. Heating 50.0 g of calcium carbonate produced 25.2 g of calcium oxide. Find the percentage yield and the atom economy.
   $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$

2. Theoretical yield: $n = 50.0 \div 100.1 = 0.4995 \text{ mol}$, so the most CaO possible is $0.4995 \times 56.1 = 28.02 \text{ g}$.

3. $$\text{yield} = \frac{25.2}{28.02} \times 100 = 89.9\%$$

4. $$\text{atom economy} = \frac{56.1}{100.1} \times 100 = 56.0\%$$

   Even at 100% yield, 44% of the mass of the limestone leaves as carbon dioxide. Yield measures how well the practical work went; atom economy measures how wasteful the reaction is.

MARK-SCHEME WORDING

- "Quoting or using the atom economy formula" is a mark on its own. Then the total of the products and the answer to 2 or 3 s.f.
- Atom economy must match the equation (use 5 × Na_2O if the equation has 5).
- Missing "× 100" is penalised only once in a paper.
- % yield: actual ÷ theoretical × 100; mass or mole routes both accepted.

Often asked as: atom economy of a given route; MCQ "which route has the highest atom economy"; % yield of a salt preparation (including hydrates and double salts); scaling up the mass needed when the yield is below 100%.

COMMON MISTAKE

- Forgetting the coefficient when a product appears more than once.
- Using the A_r of an element as the M_r of its compound (e.g. Ge, 72.6, for GeH_4).
- Dividing instead of multiplying when scaling up for a low yield.
- Quoting too many s.f.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- The M_r of a hydrated product ($MgSO_4 \cdot 7H_2O$) in a yield calculation (WCH11 Jan 2024).
- Double salts, e.g. ammonium iron(II) sulfate, with identifying the limiting reagent (WCH11 Oct 2025).
- Addition polymerisation has 100% atom economy (WCH11 Oct 2020).

SPEC 1.10 Determining a formula or confirming an equation by experiment

WHAT THIS SPEC POINT ASKS

Find a formula, or confirm an equation, from experimental results, and evaluate the data.

Typical experiments

Experiment	What you measure	What you work out
Heat a hydrated salt (e.g. $MgSO_4 \cdot xH_2O$)	mass before and after heating to **constant mass**	x, the water of crystallisation
Reduce copper oxide in a stream of hydrogen or methane	mass of oxide and of copper left	formula of the oxide
Burn magnesium in a crucible with a lid	mass of Mg and of MgO	formula MgO
Li or Mg with water or acid; collect H_2	mass of metal and volume of gas	mole ratio, so the equation

WORKED EXAMPLE

Worked example: formula of copper oxide

1.59 g of copper oxide is reduced by hydrogen, leaving 1.27 g of copper.

Mass of O = 1.59 − 1.27 = 0.32 g

Cu: 1.27 ÷ 63.5 = 0.0200 mol; O: 0.32 ÷ 16.0 = 0.020 mol → ratio 1 : 1 → **CuO**

WORKED EXAMPLE

Worked example: confirming an equation

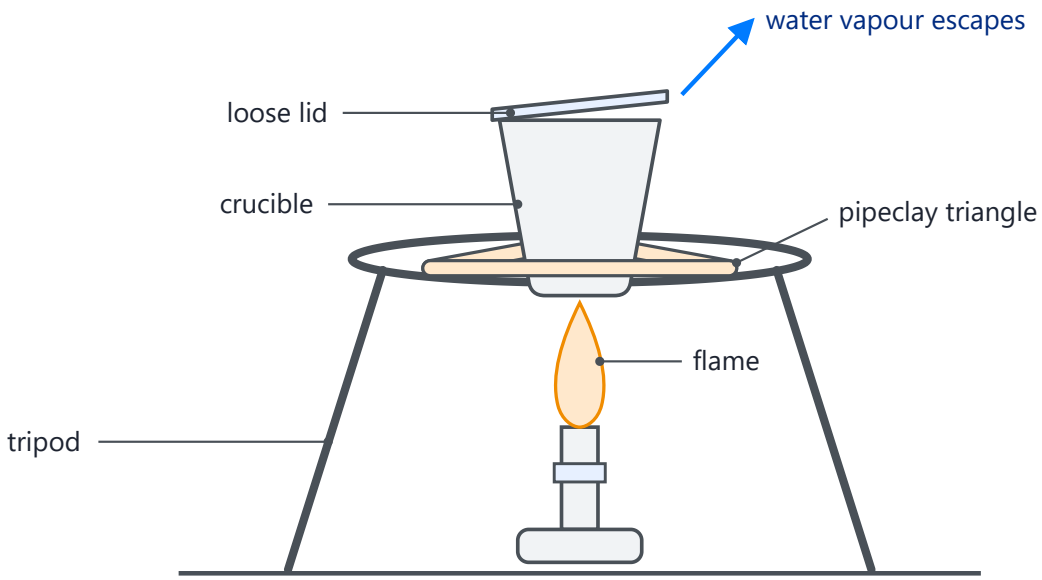
0.0350 g of lithium reacts with water, giving 60.0 cm³ of hydrogen at r.t.p.

Li: 0.0350 ÷ 6.9 = 5.07 × 10^{-3} mol; H_2: 60.0 ÷ 24 000 = 2.50 × 10^{-3} mol

Li : H_2 = 2.03 : 1 ≈ 2 : 1, which supports **$2Li(s) + 2H_2O(l) \rightarrow 2LiOH(aq) + H_2(g)$**

Evaluating the data

- **Heating to constant mass**: heat, cool, weigh, and repeat until two readings agree. This makes sure **all** the water (or other volatile product) has gone.
- **Percentage uncertainty** = (uncertainty ÷ measured value) × 100. A reading's uncertainty is half the smallest division; a mass or volume **by difference** uses two readings, so double it.
- Example: on a 2 d.p. balance each reading is ±0.005 g, so a mass by difference is ±0.01 g. For 0.32 g of oxygen that is 0.01 ÷ 0.32 × 100 = **3.1%**, much bigger than for the 1.27 g of copper (0.8%). Small masses give big percentage errors: use more sample.
- Other errors: the solid **decomposes further** (e.g. a sulfate losing SO_3) or **spits** out of the crucible; Mg **reacts with nitrogen** in air; copper **re-oxidises** if it cools in air (keep the gas flowing until cool); gas escapes or dissolves before it is measured (1.11).
- Graphs: plot one quantity against the other for several runs. A straight line through the origin confirms a fixed ratio, and its gradient gives the ratio.



A crucible on a pipeclay triangle. The lid stops solid spitting out, but it sits loosely so that water vapour can escape.

MARK-SCHEME WORDING

- "Heat to constant mass / until the mass does not change" and "to ensure all the water is lost". ALLOW "heat for longer". IGNORE repeats, a higher temperature or more solid. "Heat under reflux" is not awarded.
- Waters of crystallisation are given as **whole numbers**.
- Graph: axes labelled with units, points plotted to within half a small square, one ruled line of best fit.

Often asked as: hydrate calculations (masses or a graph); how to improve the method "using the same apparatus"; equation or formula from given masses and gas volumes; why a reactant is added in excess.

COMMON MISTAKE

- Answering "repeat and take a mean" when the question asks how to improve a single measurement with the same apparatus. The answer is usually **heat to constant mass**.
- Putting the masses of anhydrous salt and water the wrong way round.
- Naming evaporation or distillation to remove an excess solid metal: use **filtration**.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Adding magnesium in excess so all the acid reacts, then filtering off the leftover metal (WCH11 Jan 2024).
- Deducing x in a hydrate from the ratio of decomposition products, e.g. $SO_2 : SO_3 : H_2O = 1 : 1 : 14$ (WCH11 Jun 2021).
- Interpolating on a graph of mass of anhydrous salt against mass of hydrate (WCH11 Oct 2024).

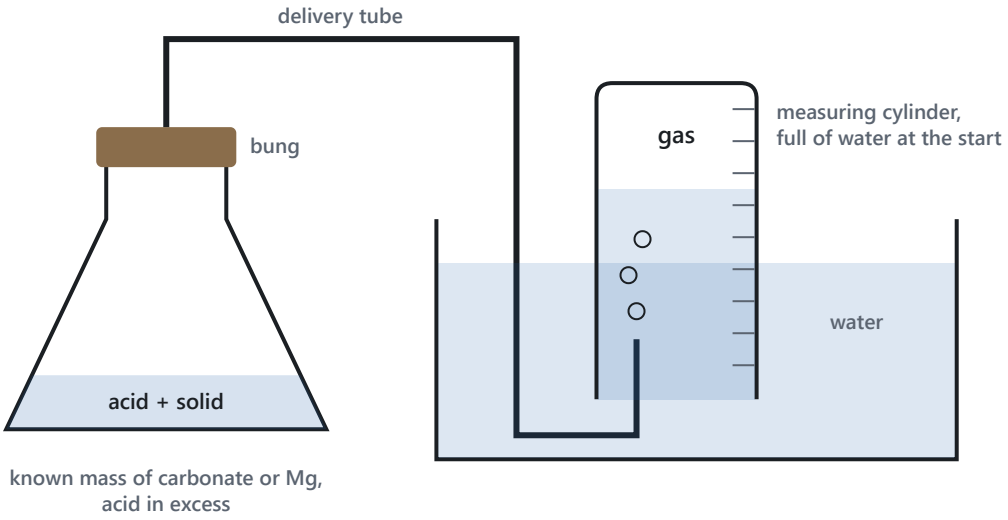
SPEC 1.11 Core Practical 1: measuring the molar volume of a gas

WHAT THIS SPEC POINT ASKS

Carry out an experiment to measure the molar volume of a gas.

Principle

Measure the volume of gas made by a **known number of moles** of solid. Molar volume = volume of gas ÷ moles of gas. A carbonate ($CaCO_3$ or $MgCO_3$) with acid gives CO_2; magnesium ribbon with acid gives H_2.



Collecting gas over water. A gas syringe in place of the measuring cylinder avoids the gas dissolving in water.

CORE PRACTICAL

Core practical 1: method in brief

1. Weigh a small sample of the solid (e.g. about 0.05–0.25 g of $MgCO_3$) on a 2 d.p. (or 3 d.p.) balance.
2. Put a measured volume of acid in the conical flask. The acid must be **in excess** so all the solid reacts.
3. Fill a measuring cylinder (or burette) with water, turn it upside down in a trough of water, and put the end of the delivery tube under it. (Or connect a gas syringe.)
4. Add the solid, **put the bung in straight away**, and collect the gas until no more bubbles appear.
5. Read the volume of gas, with your eye level with the water level in the cylinder.
6. Repeat with different masses of solid. Plot **volume of gas** (y-axis) against **mass of solid** (x-axis).

WORKED EXAMPLE

Worked example: molar volume

0.200 g of $MgCO_3$ (M = 84.3) gives 52.0 cm³ of CO_2.

$MgCO_3 + 2HCl \rightarrow MgCl_2 + CO_2 + H_2O$, so moles of CO_2 = moles of $MgCO_3$ = 0.200 ÷ 84.3 = 2.37×10^{-3} mol

Molar volume = 52.0 ÷ 2.37×10^{-3} = 21 900 cm³ mol⁻¹ = **21.9 dm³ mol⁻¹**

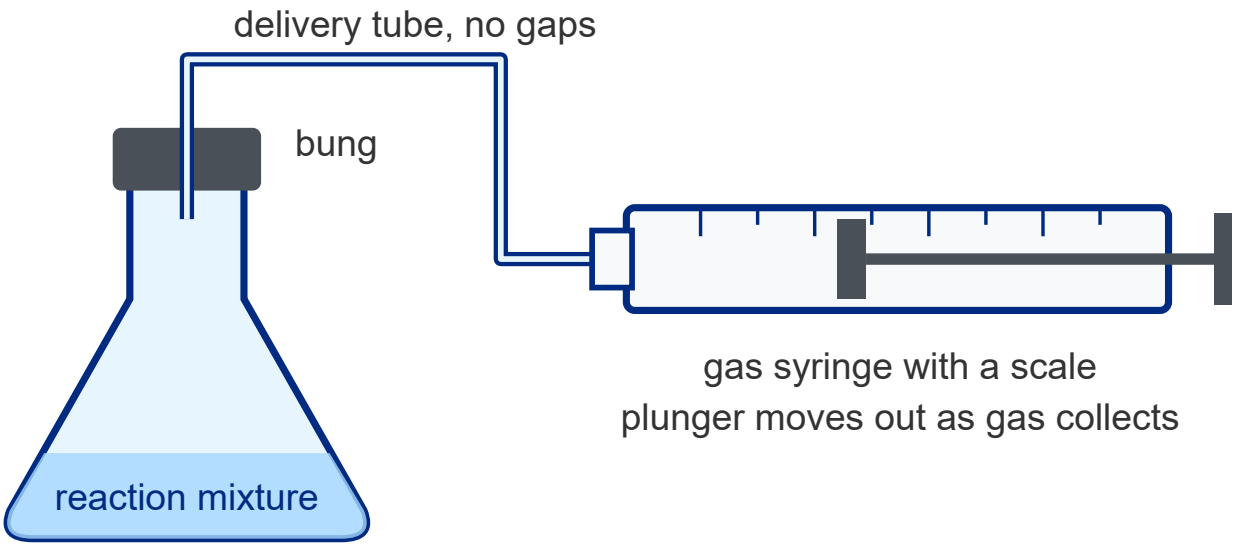
From a graph: if the gradient is 260 cm³ g⁻¹, molar volume = 260 × 84.3 = 21 900 cm³ mol⁻¹.

The textbook value at r.t.p. is 24.0 dm³ mol⁻¹, so this result is (24.0 − 21.9) ÷ 24.0 × 100 = 8.7% low.

Why the result is often too low, and how to fix it

Problem	Effect	Improvement
CO_2 dissolves in the water in the trough	less gas collected	use a **gas syringe**, or collect H_2 (from Mg) which is much less soluble
Gas escapes before the bung is put in	less gas collected	keep the solid in a small tube inside the flask, bung the flask, then tip it over to mix
Gas leaks from joints	less gas collected	check the bung and tubing are tight
Small volume read on a large cylinder	large % uncertainty	use a smaller cylinder or a burette, or a bigger mass of solid
Temperature or pressure not at r.t.p.	molar volume differs from 24.0	record them, or correct with pV = nRT

- **Variables**: independent = mass of solid; dependent = volume of gas; control = volume and concentration of acid (in excess), temperature.
- **Safety**: eye protection; dilute acids are irritants.
- **Percentage uncertainty**: reading a 100 cm³ cylinder to ±0.5 cm³ for 52.0 cm³ is 0.5 ÷ 52.0 × 100 ≈ 1%.



Collecting a gas in a gas syringe. The gas volume is read from the scale once the reaction stops; record the room temperature and pressure as well, because the volume depends on both.

MARK-SCHEME WORDING

- Graph: axes labelled with units, points on both sides of one ruled straight line, which should pass through the origin.
- Molar volume: moles of solid = mass ÷ *M*; molar volume = **your measured volume** ÷ moles, with units and 2 or 3 s.f. (about 21.5 to 22 dm³ mol⁻¹ in the WCH11 Jun 2019 data).
- Result lower than expected: "carbon dioxide **dissolves** in water", with a reason. "Leaks" must say how gas was lost. Vague "incomplete reaction", "impurities" or "reading errors" were ignored.

Often asked as: plot volume against mass and read a value off the line; calculate the molar volume; minimum concentration of acid needed (1.5); explain why the value is lower than 24 dm³ mol⁻¹.

COMMON MISTAKE

- Using 24 dm³ mol⁻¹ in the calculation when the question wants **your own** molar volume from the data.
- Reading an upside-down measuring cylinder the wrong way (the scale runs downwards).
- Joining the dots instead of drawing a line of best fit; ignoring the origin.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Reading values from a gradient or table in place of the student's own graph (WCH11 Jun 2019).
- The same set-up used for other reactions, such as Mg with sulfuric acid in a salt preparation (WCH11 Jan 2024).

SPEC 1.12 Equations from simple test-tube reactions

WHAT THIS SPEC POINT ASKS

Link ionic and full equations, with state symbols, to what you see in simple test-tube experiments: (i) displacement reactions, (ii) typical reactions of acids, (iii) precipitation reactions.

Observations are what you **see**, **hear** or **feel**: colour changes, bubbles (effervescence), a solid appearing or disappearing, the tube getting warm. Naming the product ("hydrogen is made") is an inference, not an observation.

(i) Displacement

A more reactive metal pushes a less reactive metal out of its compound; a more reactive halogen pushes a less reactive halogen out of a halide.

Reaction	Ionic equation	Observations
Zn in $CuSO_4(aq)$	$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$	blue colour fades; pink-brown solid forms; gets warm
Cu in $AgNO_3(aq)$	$Cu(s) + 2Ag^{+}(aq) \rightarrow Cu^{2+}(aq) + 2Ag(s)$	colourless solution turns **blue**; **grey/silver** solid forms on the copper
Mg in $FeSO_4(aq)$	$Mg(s) + Fe^{2+}(aq) \rightarrow Mg^{2+}(aq) + Fe(s)$	pale green colour fades; dark grey solid
$Cl_2(aq)$ + $KBr(aq)$	$Cl_2(aq) + 2Br^{-}(aq) \rightarrow 2Cl^{-}(aq) + Br_2(aq)$	colourless → **yellow-orange** (orange in cyclohexane)
Cl_2 or Br_2 + $KI(aq)$	$Br_2(aq) + 2I^{-}(aq) \rightarrow 2Br^{-}(aq) + I_2(aq)$	→ **brown** solution (**purple** in cyclohexane)

(ii) Typical reactions of acids

Reaction	Ionic equation	Observations
metal: $Mg + 2HCl \rightarrow MgCl_2 + H_2$	$Mg(s) + 2H^{+}(aq) \rightarrow Mg^{2+}(aq) + H_2(g)$	**bubbles / effervescence**; metal gets smaller and disappears; warm
carbonate: $CaCO_3 + 2HCl \rightarrow CaCl_2 + CO_2 + H_2O$	$CaCO_3(s) + 2H^{+}(aq) \rightarrow Ca^{2+}(aq) + CO_2(g) + H_2O(l)$	effervescence; solid disappears
soluble carbonate in solution	$CO_3^{2-}(aq) + 2H^{+}(aq) \rightarrow CO_2(g) + H_2O(l)$	effervescence
hydrogencarbonate	$HCO_3^{-}(aq) + H^{+}(aq) \rightarrow CO_2(g) + H_2O(l)$	effervescence
metal oxide: $CuO + H_2SO_4 \rightarrow CuSO_4 + H_2O$	$CuO(s) + 2H^{+}(aq) \rightarrow Cu^{2+}(aq) + H_2O(l)$	**black** solid dissolves (on warming); **blue** solution forms
alkali: $NaOH + HCl \rightarrow NaCl + H_2O$	$H^{+}(aq) + OH^{-}(aq) \rightarrow H_2O(l)$	no visible change; temperature rises

- Acid + metal is a **redox** reaction (the metal displaces hydrogen from the acid). Acid + base, acid + metal oxide and acid + carbonate are **neutralisations**.
- Test the gases: H_2 gives a squeaky pop with a lighted splint; CO_2 turns limewater milky.

(iii) Precipitation

Two solutions mix and an **insoluble** product appears as a solid (a precipitate).

Precipitate	Ionic equation	Colour
silver chloride	$Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$	white
silver bromide	$Ag^+(aq) + Br^-(aq) \rightarrow AgBr(s)$	cream
silver iodide	$Ag^+(aq) + I^-(aq) \rightarrow AgI(s)$	yellow
barium sulfate	$Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$	white
copper(II) hydroxide	$Cu^{2+}(aq) + 2OH^-(aq) \rightarrow Cu(OH)_2(s)$	blue
iron(II) hydroxide	$Fe^{2+}(aq) + 2OH^-(aq) \rightarrow Fe(OH)_2(s)$	green
iron(III) hydroxide	$Fe^{3+}(aq) + 3OH^-(aq) \rightarrow Fe(OH)_3(s)$	orange-brown
magnesium hydroxide	$Mg^{2+}(aq) + 2OH^-(aq) \rightarrow Mg(OH)_2(s)$	white

Solubility rules (so you know which product is (s) and which is (aq)):

- **Soluble**: all Group 1 and ammonium compounds; all nitrates; most chlorides, bromides and iodides (**not** silver or lead); most sulfates (**not** barium or lead; calcium sulfate only slightly).
- **Insoluble**: most carbonates and hydroxides (except Group 1 and ammonium; $Ca(OH)_2$ is slightly soluble).

MARK-SCHEME WORDING

- Observations, not explanations: "colourless solution turns blue"; "grey / silver solid forms"; "bubbles / effervescence" for a gas.
- Not credited: "fizzing" when no gas forms; "white precipitate" of magnesium sulfate (it is soluble); naming the gas instead of describing the bubbles.

Often asked as: copper in silver nitrate (observations plus ionic equation); Mg with acid; $CuCO_3$ with sulfuric acid; precipitating AgCl; MCQ "which type of reaction is this?" ($Zn + H_2SO_4$ is a displacement and a redox reaction).

COMMON MISTAKE

- Mixing up observation and inference: "hydrogen gas is given off" is an inference; "bubbles" is the observation.
- Wrong charge on a metal ion in solution (Cu^{2+}, not Cu^+; Ag^+, not Ag^{2+}).
- Writing (s) for a soluble salt such as $MgSO_4$, or (aq) for a precipitate.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Using the reactivity series to predict whether displacement happens, e.g. copper is more reactive than silver (WCH11 Jun 2024).
- Deciding whether a product is (aq) or (s) from solubility (WCH11 Jan 2024).
- Ionic equation for an unfamiliar precipitate, such as a basic nickel carbonate (WCH11 Oct 2023).

Quick checks

QUICK CHECK

How many oxygen atoms are there in 0.250 mol of carbon dioxide?

- 1.51 × 10^{23}
- 3.01 × 10^{23}
- 6.02 × 10^{23}
- 4.52 × 10^{23}

Each CO_2 molecule has two oxygen atoms, so there are $2 \times 0.250 = 0.500$ mol of O atoms, which is $0.500 \times 6.02 \times 10^{23} = 3.01 \times 10^{23}$. The answer 1.51 × 10^{23} counts molecules, not oxygen atoms.

QUICK CHECK

A compound of nitrogen and oxygen contains 1.40 g of N and 3.20 g of O. Its M_r is 92.0. What is its molecular formula?

- NO_2
- N_2O
- N_2O_4
- NO

N: $1.40/14.0 = 0.100$; O: $3.20/16.0 = 0.200$. The ratio 1 : 2 gives the empirical formula NO_2, with mass 46.0. $92.0/46.0 = 2$, so the molecular formula is N_2O_4. NO_2 is only the empirical formula.

QUICK CHECK

What is the concentration of chloride ions in 0.0500 mol dm^{-3} magnesium chloride solution?

- 0.0250 mol dm^{-3}
- 0.0500 mol dm^{-3}
- 0.150 mol dm^{-3}
- 0.100 mol dm^{-3} ✓

$MgCl_2 \rightarrow Mg^{2+} + 2Cl^-$, so every mole of the salt gives two moles of chloride ions: $2 \times 0.0500 = 0.100$ mol dm^{-3}. 0.150 counts all the ions (Mg^{2+} as well).

QUICK CHECK

What is the ionic equation for the reaction between sodium carbonate solution and hydrochloric acid?

- $CO_3^{2-}(aq) + 2H^+(aq) \rightarrow H_2O(l) + CO_2(g)$ ✓
- $Na_2CO_3(aq) + 2H^+(aq) \rightarrow 2Na^+(aq) + H_2O(l) + CO_2(g)$
- $2Na^+(aq) + 2Cl^-(aq) \rightarrow 2NaCl(aq)$
- $CO_3^{2-}(aq) + H^+(aq) \rightarrow CO_2(g) + OH^-(aq)$

Sodium carbonate is dissolved, so it is written as ions, and Na^+ and Cl^- are spectators. The second option keeps the dissolved sodium carbonate as a formula and leaves spectator Na^+ in the equation.

QUICK CHECK

What volume does 0.100 mol of an ideal gas occupy at 300 K and 100 kPa? (R = 8.31 J mol^{-1} K^{-1})

- 2.49 m^3
- 0.224 dm^3
- 2.49 dm^3
- 24.9 dm^3

$V = nRT/p = 0.100 \times 8.31 \times 300/100\,000 = 2.49 \times 10^{-3}\ \text{m}^3 = 2.49\ \text{dm}^3$. Leaving the pressure in kPa gives 2.49 m^3, a thousand times too big.

QUICK CHECK

Methyl orange is added to hydrochloric acid in a conical flask, and sodium hydroxide solution is run in from a burette. What colour change marks the end point?

- yellow to orange
- colourless to pink
- red to yellow
- red to orange

Methyl orange is red in acid. The end point is the first orange colour. If the solution has gone yellow you have overshot and added too much alkali. Colourless to pink is phenolphthalein.

QUICK CHECK

Which reaction has an atom economy of 100% for the product shown in bold?

- $CaCO_3 \rightarrow \mathbf{CaO} + CO_2$
- $C_2H_4 + Br_2 \rightarrow \mathbf{C_2H_4Br_2}$
- $Zn + CuSO_4 \rightarrow ZnSO_4 + \mathbf{Cu}$
- $CH_4 + H_2O \rightarrow CO + \mathbf{3H_2}$

An addition reaction has a single product, so every atom of the reactants ends up in it. All the others make a second substance that is not wanted, so their atom economy is below 100% even if the yield is perfect.

QUICK CHECK

A student heats a hydrated salt but does not heat it to constant mass before calculating x, the number of water molecules in the formula. How does this affect the value of x?

- It has no effect, because the mass of the salt is unchanged.
- x is too low, because some water is still in the solid.
- x is too high, because the anhydrous salt seems too heavy.
- x is too high, because extra water is lost.

Water left in the solid makes the "anhydrous" mass too big and the mass of water lost too small. Fewer moles of water per mole of salt means x comes out too low.